

SUDONG LEE

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EDUCATION

Ph.D. student in Robotics, Control, and Intelligent Systems (EDRS) <i>EPFL (École Polytechnique Fédérale de Lausanne)</i> <i>Advisor: Prof. Josie Hughes</i>	2023.04. - present <i>Lausanne, Switzerland.</i>
M.S. in Mechanical Engineering <i>Seoul National University</i> <i>Advisor: Prof. Yong-Lae Park</i> <i>Thesis: Modularized Robotic Skin Sensorized by Fiber Optic Force Sensing for Remote and Autonomous Robot Operation</i>	2019.03. - 2021.08. <i>Seoul, Korea.</i>
B.S. in Mechanical Engineering <i>Korea University</i>	2013.03. - 2019.02. <i>Seoul, Korea.</i>

RESEARCH EXPERIENCE

CREATE Lab (Computational design of Robots and Embodied Autonomous TEchnologies Laboratory) - <i>EPFL</i> Assistant-doctorant	<i>Lausanne, Switzerland.</i> 2023.04. - present
Soft Robotics Research Center (SRRC) - <i>Seoul National University</i> Research Associate Research Assistant Research topics: · Fiber Jamming Actuator driven by Tendons to Enhance Adaptability · Robotic Skin using 3-DoFs Force Sensor for Dexterous and Safe Interaction	<i>Seoul, Korea.</i> 2022.09. - 2023.03. 2021.09. - 2022.08.
Soft Robotics and Bionics Laboratory (SRBL) - <i>Mechanical Engineering, Seoul National University</i> Graduate Student Researcher Research topics: · Robotic Skin Sensorized by Fiber Optic Strain Sensors · Multi-modal Locomotion and Environmental Adaptability of Legged Robots · Soft Electronics and Sensors using Stretchable Materials and Sensing Mechanisms	<i>Seoul, Korea.</i> 2019.01. - 2021.08.

HONORS AND AWARDS

M.S. Thesis Presentation Award Mechanical Engineering, Seoul National University	2021.06.
Third Place Award for Locomotion Challenge IEEE International Conference on Soft Robotics 2019 (RoboSoft 2019) Team SRBL (Sudong Lee, G. Shin, J. Kim, M. Choi, Y. Baek, and Y.-L. Park)	2019.04.
Great Honor, Winter 2018 Graduation Korea University	2019. 02.
Semester High Honors Korea University	1 st Semester, 2018., 2 nd Semester, 2017., 1 st Semester, 2015., 2 nd Semester, 2014., 1 st Semester, 2014., 2 nd Semester, 2013., 1 st Semester, 2013.

SCHOLARSHIPS

Kwanjeong Fellowship
Kwanjeong Educational Foundation

2nd Semester, 2020.,
1st Semester, 2020.,
2nd Semester, 2019.,
1st Semester, 2019.

National Science and Engineering Scholarship
Korea Student Aid Foundation

2nd Semester, 2018.,
1st Semester, 2018.,
2nd Semester, 2017.,
1st Semester, 2015.

Academic Excellence Scholarship
Korea University

2nd Semester, 2014.

Best Honor Scholarship
Korea University

1st Semester, 2014.

PUBLICATIONS

Journal Papers

1. **Sudong Lee**, K. Gilday, and J. Hughes, "Tendon-driven robotic shoulder joint with passive stabilization for large workspace," *IEEE/ASME Transactions on Mechatronics*, 2026. (DOI: [10.1109/TMECH.2026.3674717](https://doi.org/10.1109/TMECH.2026.3674717))
2. A. Georgopoulou, M. A. Calvo, L. Lucherini, **Sudong Lee**, J. Hughes, and E. Amstad, "Programmable somatosensory soft robots," *npj Flexible Electronics*, vol. 10, 58, 2026. (DOI: [10.1038/s41528-026-00558-0](https://doi.org/10.1038/s41528-026-00558-0))
3. A. Georgopoulou, **Sudong Lee**, B. Dai, F. Bono, J. Hughes, and E. Amstad, "3D printing of self-healing longevous multi-sensory e-skin," *Communications Materials*, vol. 6, 121, 2025. (DOI: [10.1038/s43246-025-00839-7](https://doi.org/10.1038/s43246-025-00839-7))
4. **Sudong Lee***, J. I. Kim*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, "Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control," *IEEE Transactions on Robotics*, vol. 40, pp. 2373-2389, 2024. (DOI: [10.1109/TRO.2024.3378178](https://doi.org/10.1109/TRO.2024.3378178))
*: **Sudong Lee and J. I. Kim are co-first authors.**
5. D. Kim, **Sudong Lee**, T. H. Hong, and Y.-L. Park, "Exploration-based model learning with self-attention for risk-sensitive robot control," *npj Robotics*, vol. 1, 7, 2023. (DOI: [10.1038/s44182-023-00006-5](https://doi.org/10.1038/s44182-023-00006-5))
6. J. Kang*, **Sudong Lee***, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7344-7351, 2023. (DOI: [10.1109/LRA.2023.3316075](https://doi.org/10.1109/LRA.2023.3316075))
*: **Sudong Lee and J. Kang are co-first authors.**
7. T. Kim*, **Sudong Lee***, S. Chang, S. Hwang, and Y.-L. Park, "Environmental adaptability of legged robots with cutaneous inflation and sensation," *Advanced Intelligent Systems*, vol. 5, no. 11, 2300172, 2023. (DOI: [10.1002/aisy.202300172](https://doi.org/10.1002/aisy.202300172))
*: **Sudong Lee and T. Kim are co-first authors.**
Cover Article: [10.1002/aisy.202370050](https://doi.org/10.1002/aisy.202370050), Editor's Choice: [\[Link\]](#)
8. Y. Lee, S. Lim, W. J. Song, **Sudong Lee**, S. J. Yoon, J.-M. Park, M.-G. Lee, Y.-L. Park, and J.-Y. Sun, "Triboresistive touch sensing: grid-free touch-point recognition based on monolayered ionic power generators," *Advanced Materials*, vol. 34, no. 19, 2108586, 2022. (DOI: [10.1002/adma.202108586](https://doi.org/10.1002/adma.202108586))
9. G. Shin*, **Sudong Lee***, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4983-4990, 2022. (DOI: [10.1109/LRA.2022.3153707](https://doi.org/10.1109/LRA.2022.3153707))
*: **Sudong Lee and G. Shin are co-first authors.**
10. T. Kim, **Sudong Lee**, T. Hong, G. Shin, T. Kim, and Y.-L. Park, "Heterogeneous sensing in a multifunctional soft sensor for human-robot interfaces," *Science Robotics*, vol. 5, no. 49, eabc6878, 2020. (DOI: [10.1126/scirobotics.abc6878](https://doi.org/10.1126/scirobotics.abc6878))

Conference Papers

1. **Sudong Lee** and J. Hughes, "Morphological and material programability of a hall-effect based soft tactile sensors," *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, San Diego, CA, USA, 2024, pp. 325-331. (DOI: [10.1109/RoboSoft60065.2024.10521990](https://doi.org/10.1109/RoboSoft60065.2024.10521990))

CONFERENCE PRESENTATIONS

1. A. Georgopoulou, **Sudong Lee**, B. Dai, J. Hughes, and E. Amstad, "Multimodal selective sensory receptors for robotic e-skin," *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)*, Lausanne, Switzerland, 2025.
2. **Sudong Lee***, J. I. Kim*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, "Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control," *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Abu Dhabi, UAE, 2024.
*: **Sudong Lee and J. I. Kim are co-first authors.**
3. J. Kang*, **Sudong Lee***, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *2024 IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024.
*: **Sudong Lee and J. Kang are co-first authors.**
4. G. Shin*, **Sudong Lee***, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *2022 IEEE 5th International Conference on Soft Robotics (RoboSoft)*, Edinburgh, United Kingdom, 2022.
*: **Sudong Lee and G. Shin are co-first authors.**

PATENTS

1. T. Kim, **Sudong Lee**, and Y.-L. Park, "Soft Sensor with Multi-Sensing Function," 2022.
(Korea Patent: 10-2384623)

TEACHING EXPERIENCE

Assistive Technology Challenge (ATC) - HACKAHEALTH, EPFL Teaching assistant, Instructor: Prof. Auke J. Ijspeert.	Spring semester, 2024-25.
ME-320: Product development and engineering design - Mechanical Engineering, EPFL Teaching assistant, Instructor: Prof. Josie Hughes.	Autumn semester, 2024-25.
Interdisciplinary robot competition - EPFL Team coach, Instructor: Prof. Auke J. Ijspeert.	Spring semester, 2023-24.
ME-320: Product development and engineering design - Mechanical Engineering, EPFL Teaching assistant, Instructor: Prof. Josie Hughes.	Autumn semester, 2023-24.
M2794.001700_001: Mechanical product design - Mechanical Engineering, Seoul National University Teaching assistant, Instructor: Prof. Yong-Lae Park.	1 st semester, 2019.

ACADEMIC SERVICE

Reviewer - Journal Papers <i>IEEE Transactions on Robotics (T-RO)</i> <i>IEEE/ASME Transactions on Mechatronics (T-MECH)</i> <i>IEEE Transactions on Industrial Electronics (TIE)</i> <i>IEEE Robotics and Automation Letters (RA-L)</i> <i>Nature Communications</i> <i>Soft Robotics</i>
Reviewer - Conference Papers <i>IEEE International Conference on Robotics and Automation (ICRA)</i> <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> <i>IEEE International Conference on Soft Robotics (RoboSoft)</i>

TECHNICAL STRENGTHS (SKILLS)

Programming Languages	C++, Python, Matlab
Embedded System	Arduino, AVR ATmega, Single-Board Computer (SBC)
Software for System and Robots	ROS, Pybullet
Machine Learning	Pytorch, TensorFlow
Design and Simulation	3D Computer-Aided Design (CAD), Finite Element Analysis (FEA) Software
Fabrication	3D Printing (Additive Manufacturing), Silicone Fabrication

OTHER EXPERIENCE

Republic of Korea Air Force (ROKAF, Military Service)
Staff Sergeant, Honorable discharge.

2015.08. - 2017.08.